

Flying uncharted skies: Unravelling the network of contributory factors to adverse Australian RPAS events with an aggregate AcciMap

Samantha L. Jackson^{a,*}, Paul M. Salmon^a, Gemma J.M. Read^{a,b}

^a Centre for Human Factors and Sociotechnical Systems, University of the Sunshine Coast, Australia

^b School of Health, University of the Sunshine Coast, Australia

ARTICLE INFO

Keywords:

Systems thinking

AcciMap

Remotely piloted aircraft systems (RPAS)

Uncrewed aerial systems (UAS)

Uncrewed aerial vehicles (UAV)

Accident analysis

ABSTRACT

Remotely Piloted Aircraft Systems (RPAS) are a rapidly expanding technology that operates within the larger complex aviation system. As a result, protective frameworks and risk controls for supporting safe operation are still developing. Adverse events are occurring, yet it is unclear what systemic factors interact to create them. To identify vulnerabilities in the broader Australian RPAS network, this study used AcciMap to analyse a set of adverse RPAS events investigated by the Australian Transport Safety Bureau. Investigation reports into RPAS incidents in Australia ($n = 14$) were identified and analysed in-depth with AcciMap; individual AcciMaps were constructed for each adverse event. The data was then aggregated to create a network of contributory factors across the RPAS network in Australia. The findings revealed 232 individual contributory factors across five of the six AcciMap levels. The contributory factors were then coded using a pre-existing classification scheme to identify 31 contributory factor themes. The highest proportion of contributory factors were at the lower operational and equipment levels of the system; these factors included *Equipment, technology and resources*; *Activity, work and operations*; and *Compliance with procedures, violations and unsafe acts*. According to systems theory and previous analyses using AcciMap, we would expect to find organisational, regulatory and government contributory factors, yet such factors were limited in this sample. Further research is required to explore the factors across the entire sociotechnical system that contribute to adverse RPAS events.

1. Introduction

Remotely Piloted Aircraft Systems (RPAS) are rapidly developing technologies that consist of many networked parts, including the remotely piloted aircraft (RPA) itself (also known as an uncrewed aerial vehicle, UAV), ground-based control station(s) and communication systems (Stewart and Martin, 2021). These technologies can range in size from small, inexpensive, hand-launched vehicles to large multi-person operated systems with powerful engines and extended endurance capability (Cooke et al., 2017; Hobbs, 2010). RPAS have been adopted across military, commercial, and consumer applications for various functions, including photography, surveying, search and rescue, and warfare operations (Cooke et al., 2017; Fortune Business Insights, 2023; Stewart and Martin, 2021). In 2022, the global market for RPAS was estimated at USD \$27.43 billion; by 2030, this is expected to expand to USD \$91.23 billion (Fortune Business Insights, 2023). In Australia, projections estimate that by 2043, there will be 60.4 million RPA flights

(which consist of one take-off and landing) each year. This represents a 20% increase annually from current figures for the next 20 years (Scyne Advisory, 2024).

Both RPAS and the broader aviation environments in which they operate can be conceptualised as complex systems. They include multiple humans and technologies working together toward a shared goal, their boundaries are hard to define, and interactions are non-linear, meaning outcomes are difficult to predict. Many subsystems within the larger RPAS interact as dynamic elements that are managerially and/or operationally independent. These subsystems often interact within and between other subsystems, demonstrating complex behaviours (Walden et al., 2015). Systems theory is a valuable framework for understanding and enhancing complex systems, with a focus on examining the whole system rather than its component behaviour (Cilliers, 1998; Ottino, 2003; Read et al., 2021). Examining the system in this way enables the identification of a range of contributory factors and how they interact to create incidents and other adverse events. By better

* Corresponding author. Centre for Human Factors and Sociotechnical Systems University of the Sunshine Coast 90 Sippy Downs Dr Sippy Downs, Queensland, 4556, Australia.

E-mail address: samantha.jackson@research.usc.edu.au (S.L. Jackson).

<https://doi.org/10.1016/j.apergo.2025.104468>

Received 16 September 2024; Received in revised form 12 December 2024; Accepted 18 January 2025

Available online 27 January 2025

0003-6870/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

understanding the interactions between contributory factors across an entire sociotechnical system, more systemic and, thus, more effective safety controls can be identified. Rasmussen's (1997) risk management framework (RMF), founded in systems theory, is useful for conceptualising contributory factors across a whole system (Salmon et al., 2020). Rasmussen proposed that systems consist of various hierarchical levels wherein actors (individuals and organisations) make decisions and take actions that interact across levels and ultimately lead to adverse events. The hierarchical levels utilised can vary but generally include 1) government at the highest level, followed by 2) regulators and associations, 3) company, 4) management, 5) staff, and 6) equipment and environment at the lowest level.

In any complex and dynamic system, there is a higher risk of adverse events (Perrow, 1999; Qureshi and Campbell, 2009), and several significant RPAS incidents have already occurred. In particular, this rapidly expanding technology presents a safety challenge for other aircraft (ATSB, 2017a; Clothier et al., 2018; EASA, 2021), as well as people and infrastructure on the ground (Clarke and Bennett Moses, 2014; Clothier et al., 2018; Johnson et al., 2019). An example of such a safety challenge is the well-known incident at Gatwick Airport in the United Kingdom in December 2018. In this example, alleged sightings of an RPA(s) were reported over two days, leading to the cancellation or diversion of more than 1000 flights due to the risk of collision (Shackleton, 2020). Official bodies have also investigated many episodes of near-misses or collisions involving RPAs (Pyrgies, 2019). One recent case investigated by the Australian Transport Safety Bureau (ATSB) was the in-flight breakup of an Airbus Zephyr UAV on September 28, 2019 (ATSB, 2020). The investigation found that a contributing factor to the accident was the aircraft's limitations in detecting changing atmospheric conditions. When the RPA encountered unstable conditions, several uncommanded turns led to the vehicle diverting from the planned flight and the subsequent in-flight breakup. In response, Airbus implemented improved controls to forecast weather phenomena to inform mission planning and in-flight decision-making.

Accident investigation reports provide a rich source of information about contributing factors to adverse events (ICAO, 2013; Wild et al., 2017). Given the emerging nature of RPAS and the immaturity of existing safety management systems (CASA, 2022; Neville and Williams, 2016), it is critical to understand the interactions that create adverse events, as these suggest gaps in existing protective structures. Different countries have different regulations and approaches for RPAS based on characteristics such as the system's purpose (e.g. recreational, commercial) and/or weight (Wiedemann et al., 2024). In Australia, these are considered the same from an investigative perspective. A benefit of capturing and analysing accident data relating to all types of RPAS is the comprehensive safety insights produced, which may indicate higher-level safety issues and trends that are not apparent when focusing on one category. Further, it provides a more holistic evaluation of the safety landscape when considering interventions, particularly relating to regulatory structure and policy (Wild et al., 2016).

Two analysis methods designed to analyse contributory factors from adverse events are AcciMap (Svedung and Rasmussen, 2002) and the Human Factors Analysis and Classification System (HFACS; Shappell and Wiegmann, 2001). Both methods have been identified as popular choices for accident analysis (Hulme et al., 2019; Salmon et al., 2020). HFACS has been applied to the RPAS domain several times (Giese et al., 2013; Grindley et al., 2024; Manning et al., 2004; Thompson et al., 2005; Tvaryanas et al., 2006; Tvaryanas and Thompson, 2008). In contrast, AcciMap is yet to be applied to the RPAS context despite many applications within other safety-critical domains (Hulme et al., 2019; Salmon et al., 2020; Waterson et al., 2017). A detailed comparison of these two methods is beyond the scope of this paper, but other papers (see Hulme et al., 2024; Salmon et al., 2012) have discussed the merits and drawbacks of each. Briefly, however, HFACS, underpinned by Reason's Swiss cheese model, largely focuses on human error and uses taxonomy to guide error and contributory factor classification over four levels

(Shappell and Wiegmann, 2001). The taxonomy makes it well-suited for the analysis of multiple accident cases to identify trends in the data. Whilst the use of the taxonomy improves the method's reliability, it also constrains analysts to specific failures identified through the bounds of the taxonomy. A significant limitation of HFACS is that when used alone, the method does not identify specific relationships between contributing factors, only frequencies of different failure types (Hulme et al., 2019; Lenné et al., 2012). To increase its usefulness, quantitative or statistical analyses, like odds ratios, are often added to increase the understanding of relationships and the strength of relationships (Hulme et al., 2019). Additionally, the focus of HFACS on human error is not aligned with contemporary theories of incident/accident causation and limits its ability to provide a comprehensive system-wide analysis (Read et al., 2021; Salmon et al., 2012; Zhang et al., 2018). The AcciMap method is underpinned by Rasmussen's (1997) RMF and provides an in-depth analysis and valuable insights into systemic factors contributing to adverse events. A prominent characteristic of the AcciMap method is its ability to examine relationships between contributory factors, notably distinguishing it from HFACS. Previously, there have been limitations with AcciMap because the lack of a contributory factor classification scheme makes it difficult to reliably aggregate data, but recent studies (Read et al., 2022; Salmon et al., 2017, 2020) have extended the AcciMap method by developing a classification scheme to support analysis of multiple incident datasets. This goes beyond HFACS by incorporating many more categories (79 vs 19), including factors beyond the organisation (i.e. from the decisions and actions of regulators, standards-setting bodies and governments). The process allows a comprehensive systemic analysis of contributory factors whilst applying a classification scheme to better examine systemic trends in the data. The growing use of RPAS underscores the need for a comprehensive systems analysis method. Application of the AcciMap classification scheme can help identify data trends that lead to safety interventions, essential for evolving the sociotechnical safety systems of RPAS.

Given that AcciMap and the AcciMap taxonomy have not yet been applied to RPAS incidents, this paper describes the findings from a study that used AcciMap to analyse adverse RPAS events in the Australian context. The aim was to conduct an in-depth analysis to identify the patterns of contributory factors involved in these adverse events. Relationships between contributory factors are examined, and key contributory factors are identified to reveal systemic vulnerabilities and areas for intervention.

2. Method

The AcciMap method (Svedung and Rasmussen, 2002) was applied to 14 ATSB investigation reports to analyse adverse events in-depth and identify contributory factors and their interrelations. Salmon et al.'s (2020) contributory factor classification scheme was then applied to classify the identified contributory factors and create an aggregate AcciMap covering all 14 adverse events. This enabled an in-depth analysis of contributory factor themes across the system.

2.1. AcciMap

AcciMap (Svedung and Rasmussen, 2002), developed based on Rasmussen's (1997) RMF, provides a means to identify, visualise, and describe the contributory factors involved in adverse events. This analysis consists of two key parts. The ActorMap graphically represents the stakeholders (i.e. actors) that exist at each of the six hierarchical levels of the system. The decisions and actions of these stakeholders may represent contributory factors to the adverse event, and the ActorMap provides a representation of where the stakeholders reside across system levels, supporting decisions about mapping contributory factors to different system levels. The AcciMap illustrates the contributory factors to adverse events and how they relate to each other across and within a hierarchical system. The system levels utilised in the ActorMap and

AcciMap can be modified depending on the domain, but those typically used include government policy and budgeting, regulatory bodies and associations, local area government planning and budgeting, technical and operations management, physical processes and actor activities, and equipment and surroundings. The AcciMap technique was recently extended by aggregating data over several adverse events to develop a general contributory factors classification scheme (Salmon et al., 2020). This classification scheme enables AcciMap to analyse multi-incident datasets and has been subsequently applied to aggregate contributory factors across transportation incidents and accidents (Read et al., 2022). The classification scheme comprises 79 contributory factor types across the six standard AcciMap levels (see Appendix A for the complete list of these thematic contributory factors), enabling multi-accident analysis to identify systemic causal links from the aggregated relationships (Salmon et al., 2020). Through this process patterns of recurring causal links can be identified.

2.2. Research ethics

The Research Ethics and Integrity Office of the University of the Sunshine Coast (UniSC) provided an exemption from the requirement to obtain research ethics approval based on the use of publicly available information (reference: OE23064).

2.3. Identification and selection of cases

To identify Australian RPAS cases for analysis, a structured search was conducted on the online National Aviation Occurrence Database hosted by the ATSB. The ATSB is an independent government organisation responsible for investigating adverse transport events and safety occurrences to identify and reduce safety risks (Transport Safety Investigation Act 2003, 2024). The ATSB (ATSB, n.d.) conducts various levels of investigation based on the expected scope and scale of the necessary work to identify the safety factors leading to an incident. Investigation levels are classified as Occurrence Briefs, Short Investigations, Defined Investigations, Systemic Investigations, and Major Investigations. Occurrence Briefs are short factual summaries of the occurrence and are not investigations. All other categories are considered investigations and produce reports that provide at least a summary, some analysis, and findings relating to the adverse event. Whilst Short Investigations include safety factors (i.e. factors that increase the risk of an adverse event), they do not always specify contributing factors (i.e. safety factors causally linked to an event; ATSB, n.d.). The ATSB was chosen as a source for this analysis given its application of structured and reliable investigation methodology and its transparent publication of full-length reports in English. To be included in the analysis, investigation reports needed to meet the following criteria:

1. Published between 07 Jan 03 and 17 Jan 24. This start date was selected as it was the earliest occurrence date for any investigated adverse event listed on the ATSB online database following the establishment of the ATSB as an independent Commonwealth Statutory agency in 2003 (Transport Safety Investigation Act 2003, 2024). This date also roughly coincides with the first drafting in 2001 of the regulations governing RPAS in Australia (Cho, 2013).
2. Involved circumstances where the ATSB had investigated an adverse RPAS event, resulting in a report. All investigation types, excluding occurrence briefs, were eligible for inclusion.
3. The report's status was 'final', indicating the investigation was complete. Active or discontinued investigations, as well as interim and pending reports, were excluded.

Two search strategies were used. In the first strategy, the entire occurrences database was filtered by Aircraft Type "Remotely Piloted Aircraft" and then by "investigations only". The second strategy was to use the text search filter to search the entire database with the terms "UAV"

(Uncrewed Aerial Vehicle), "Unmanned", "Uncrewed", "RPA", "Drone", and "Remotely piloted". After duplicates were removed, the search process resulted in 14 reports.

2.4. Construction of AcciMaps and coding of contributory factors and relationships

The aggregate AcciMap was constructed in two stages. A flowchart depicting this process is presented in

Fig. 1. The first stage followed the step-by-step process outlined in Salmon et al. (2022) to develop individual ActorMaps and AcciMaps for each of the 14 adverse RPAS events. The second stage involved coding contributory factors in each AcciMap using the generic AcciMap classification scheme described in Salmon et al. (2020). The authors discussed the inclusion and classification of contributory factors throughout both stages of the process, with formal inter-rater reliability

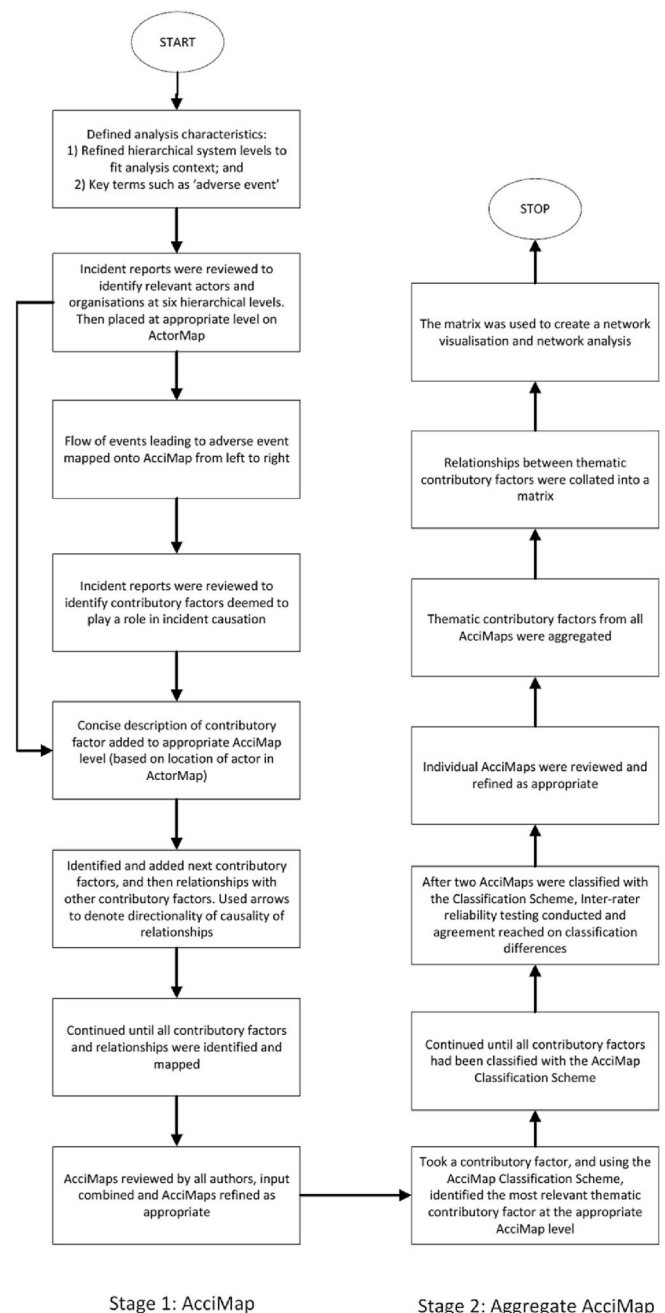


Fig. 1. Aggregate AcciMap procedure (adapted from Salmon et al., 2022).

testing performed in the second stage. A detailed explanation of the complete process is below. The authors have considerable experience applying human factors methods in research and practice. SJ is a registered psychologist with a postgraduate qualification in safety and accident analysis, has worked in Aviation for over five years, and has more than four years of experience in human factors research. GR has over 16 years of experience, and PS has over 22 years of experience applying human factors methods in research and practice, including extensive expertise in applying AcciMap and contributory factor classification schemes. GR and PS provided training to SJ in AcciMap analysis and using the classification scheme before the analysis commenced.

Before the AcciMaps were constructed, the scope of analysis was defined as extending until *the end of the adverse event*. Usually, this was the impact of the RPA into an object. However, in the case of a close encounter between RPA and crewed aircraft, it included factors immediately after the near miss to return the aircraft (either the RPA or crewed aircraft) to a safe state¹; this was considered the end of the adverse event. Data was extracted exclusively from the ATSB accident investigation reports. Given this study aimed to determine vulnerabilities and areas for intervention in the Australian RPAS network and broader aviation system, the traditional hierarchical system levels extending to the Australian government and international agencies (i.e. ICAO) were applied. These were: *Level 1 Equipment and surroundings*; *Level 2 Physical processes and actor activities*; *Level 3 Technical and operational management*; *Level 4 Government, company management, planning and budgeting*; *Level 5 Regulatory bodies and associations*; *Level 6 Government policy and budgeting*. In the first stage, one analyst (SJ) constructed ActorMaps from each report to identify the relevant actors and organisations at the six hierarchical levels. This involved reading all sections of each report, identifying all stakeholders (i.e., organisations and actors) mentioned, and then placing them at the appropriate hierarchical level in the ActorMap. Once this had been completed, all sections of the report were again reviewed to develop an AcciMap for each report. This commenced by mapping the flow of events leading to the adverse event from left to right on the AcciMap hierarchical structure. ATSB report formats have evolved over time, leading to slight variations across reports. Typically, however, information about the sequence of events was initially gathered from ‘The occurrence’ or ‘What happened’ summary sections. The information provided in these summary sections and the depth of further analysis in other sections varied between reports depending on the scope and type of ATSB investigation. After mapping the sequence of events, contributory factors were identified using the ATSB (2008) definition of a contributory safety factor, being a factor that “... that, if it had not occurred or existed at the relevant time, then 1) the occurrence would probably not have occurred; 2) adverse consequences associated with the occurrence would probably not have occurred or have been as serious; 3) another contributing safety factor would probably not have occurred or existed ...” (ATSB, 2023 p. 152). Due to the differences in scope and level of ATSB investigation reports, contributory factors were not specified as a separate section in all reports, but all reports provided sufficient contextual factual information, analysis and findings to support the identification of contributory factors. To decide if a factor met the definition of being contributory, all sections of the reports were reviewed, and the analyst applied their judgement. Examples from the report and the corresponding contributory factors are provided in Table 1 to illustrate how contributory factors were identified from the text. Once contributory factors were identified, a summary description of the factor was drafted, then matched to a relevant actor in the ActorMap (i.e. the actor that made the decision or took that action) to identify the appropriate level, and finally, the factor

Table 1
Text examples and corresponding contributing factors.

ATSB investigation report text	Contributing factor
“At the time of the incident, there was no regulation that applied to fatigue management for operators of RPA ... ” (ATSB, 2022, p.10)	No regulation for RPAS Operator Fatigue management (Level 5)
“The UAV operator broadcast on the area frequency advising his intention to conduct unmanned aerial photography operations over the Echo mine site for the next 4 h, and addressed the ‘ag aircraft operating north-east of Echo mine site’, but did not receive a response ... ” (ATSB, 2014a, p.1)	Operator broadcast on area frequency advising intent to conduct flight operations over Echo mine site, addressing ‘ag acft op NE of Echo mine site’. Operator did not receive response (Level 2)
“The pilot did not receive any alerts on the traffic collision avoidance system (TCAS) ... ” (ATSB, 2014b, p.1)	The Bell pilot did not receive any alerts on the traffic collision avoidance system (Level 1)
“The manufacturer of the airship had provided the operator with basic documentation and no training ... ” (ATSB, 2013, p.2)	Manufacturer provided operator with basic documentation and no training (Level 4)

was mapped onto the AcciMap. If stakeholders identified in the ActorMap could not be mapped to an action in the AcciMap, they were removed from the ActorMap.

As each contributory factor was identified and mapped, relationships between contributory factors were concurrently identified. Relationships did not need to be expressly stated as such in the report; analyst judgment was applied to identify if a specific contributory factor was created, influenced or occurred in a sequence of contributory factors to determine if it had a relationship with another (Salmon et al., 2022). The direction of a relationship is shown by the arrow. If one contributory factor influenced or caused another contributory factor, or if it occurred in a specific sequence, the arrowhead will point in the direction of that influence or sequence (Salmon et al., 2022). It was necessary to establish rules to ensure the replicability of the analysis. An example rule was that if a weather event was noted as simply occurring, it was placed at Level 1 Equipment and surroundings; for example, *Sunny and warm conditions, increasing risk of dehydration*. Whereas if a weather event affected a process or changed the state of the RPA, it was placed at Level 2 Physical processes and actor activities; for example, *Atmospheric conditions exceeded RPA ability to remain in controlled flight*. Similarly to the identification process for contributory factors, all sections of the report were reviewed to determine the presence of relationships between contributory factors. For example, within Report AO-2019-056 (ATSB, 2020), In-flight break-up involving Airbus Zephyr, a contributory factor was *Crew attempted to return RPA to a controlled flight but were unsuccessful* (Level 2). The unsuccessful crew action was directly related to the factor *RPA airspeed increased beyond VNE (Velocity Never Exceeded or may result in structural damage)* (Level 1) because control of the aircraft could not be regained despite direct attempts by the crew to do so. Therefore, a relationship was identified between the two. Similarly, the increase in airspeed beyond VNE then led to *RPA broke-up in-flight* (Level 2), and a relationship was documented between the two factors. Finally, while formal inter-rater reliability testing did not occur in stage one, all authors reviewed the output from each AcciMap as it was constructed to calibrate and refine the coding. Minor refinements and amendments were made during this phase, and rules were documented in an ongoing manner and applied to future analyses.

In the second stage of constructing the aggregate AcciMap, the first author (SJ) coded all contributory factors in two AcciMaps using the classification scheme developed by Salmon and colleagues (2020). The classification scheme has 79 contributory factor themes across six levels of the AcciMap. In coding, each contributory factor was considered against the thematic contributory factors at the corresponding hierarchical level in the classification scheme. For example, the contributory

¹ In some close encounter cases, the identity of the RPA and crew was unknown. Thus, the return of the crewed aircraft to a safe state was considered the end of the adverse event, and nothing further was coded for the RPA.

factor mentioned earlier of *Crew attempted to return RPA to a controlled flight but were unsuccessful* was considered against the 13 potential contributing factors in the classification scheme at Level 2, and the classification of *Activity, work and operations* was selected. To calibrate the coding using the classification scheme, the second author (PS) then conducted a formal inter-rater reliability check by independently classifying the contributory factors using the same process described above. A percentage agreement and Cohen's Kappa statistic were calculated by comparing the coded contributory factors across the two analysts. This resulted in a high level of agreement (84%, Kappa = 0.812, 95% CI 0.62 to 0.96). Again, differences between analysts were discussed, and final versions were agreed upon. The contributory factors identified in all AcciMaps were then aggregated to produce an AcciMap representing the common recurring systemic contributory factors identified from the set of investigation reports. The 'most common contributory factors' at each of the six levels were highlighted in a second, more focused AcciMap, which enabled a more detailed examination of the causal pathways between these common contributory factors. The most common contributory factors were identified if their frequency was more than 10% within the level following Salmon et al. (2020).

2.5. Network analysis

To further analyse the AcciMaps, a network representation was created, where the contributory factor themes were considered nodes within the network, and the relationships were represented by connections between nodes. Using Excel, a matrix representing the frequency of node connections between contributory factors was recorded for each AcciMap. All connections were then collated into one matrix representing the network of connections between thematic codes derived from analysing all adverse events. A network visualisation was created using the Social Network Visualizer (SocNetV) software program to show the relationships among thematic codes. Additionally, the network analysis allowed for representing relationships between factors coded to the same theme. SocNetV was then used to examine the aggregate AcciMap using two centrality metrics:

2. *Outdegree centrality*, also known as *degree centrality*. This metric provides a centrality measure indicated by the number of connections a node has outwards to other nodes in a network; this represents the degree of influence that the node projects in the network. More

connections indicate a greater influence on the network (Hanneman and Riddle, 2005; Knoke and Burt, 1983; Wasserman and Faust, 1994). With accident causation, a higher outdegree centrality indicates that the contributory factor has a greater influence in creating or influencing other contributory factors in the AcciMap (Read et al., 2022; Salmon et al., 2020).

Indegree centrality, also known as *degree prestige*. This metric provides a centrality measure indicated by the number of connections into a node from other nodes in a network; this represents how influenced a node is by other nodes within the network. More connections inbound indicate that the network influences the node more (Hanneman and Riddle, 2005; Knoke and Burt, 1983; Wasserman and Faust, 1994). In accident causation, contributory factors with higher indegree centrality exist due to or are more influenced by other contributory factors (Read et al., 2022; Salmon et al., 2020).

Key nodes were identified by calculating the mean centrality measure scores plus one standard deviation (SD) for all network nodes using the indegree and outdegree centrality scores. Key nodes are the most central agents within the network and have the greatest influence on it relative to other agents (Houghton et al., 2006; Read et al., 2018).

3. Results

A summary of the incidents analysed is presented in Table 2. The individual AcciMaps are presented in supplementary material 1.

3.1. Frequency of contributory factors

A total of 232 contributory factors were identified across five of the six AcciMap levels. The number of contributory factors varied considerably between adverse events. On average, 16.57 contributory factors per AcciMap were found (SD 9.90; range 6–46 contributory factors). The total number of contributory factors was subsequently classified into 31 contributory factors from Salmon et al.'s (2020) classification scheme. Fig. 2 shows the frequency and proportion of contributory factors identified at each level; this is represented as a proportion of the overall number of factors and the proportion each factor represents within the level. Of all levels, Level 2 of the AcciMap has the highest overall proportion of contributory factors (52.59%). The most frequent contributory factor themes within Level 2 were *Activity, work and operations*

Table 2
Reports included in the analyses.

ATSB report number	Occurrence type(s)	Occurrence date	Remotely piloted vehicle	Occurrence location	Outcome
AO-2021-001	Loss of control, Collision with terrain	15 Jan 21	DJI Inspire 2	Darling Harbour, Sydney, New South Wales	RPA destroyed
AO-2020-035	Loss of control	19 Jun 20	RF Designs Mephisto	Bruhl Airfield, Queensland	Minor damage to RPA
AO-2019-056	In-flight break-up	28 Sep 19	Airbus Zephyr 8	Near Wyndham Airport, Western Australia	RPA destroyed
AO-2019-004	Collision with terrain	09 Jan 19	Insitu ScanEagle X200	Near Woleebee Creek, Queensland	RPA destroyed
AO-2016-141	Collision with terrain	25 Oct 16	Lockheed Martin Stalker eXtended Endurance (XE)	Avoca Race Track, Victoria	Substantial damage to RPA
AO-2017-043	Collision with terrain	06 Apr 17	Yamaha RMAX	23 km W of Canberra, Australian Capital Territory	Substantial damage to RPA
AO-2016-139	Collision with terrain	24 Oct 16	Lockheed Martin Stalker eXtended Endurance (XE)	Mount Disappointment, Victoria	Substantial damage to RPA
AO-2016-128	Loss of control	27 Sep 16	Pulse Aerospace Vapor 55	4 km NE of Ballina/Byron Gateway Airport, New South Wales	Unknown (Missing)
AO-2015-112	In-flight break-up	19 Sep 15	DJI 'Spreading Wings' S900	Toowoomba, Queensland	RPA destroyed
AO-2015-035	Collision with terrain	29 Mar 15	Aeronavics SkyJib 8	Near the Melbourne Cricket Ground, Melbourne, Victoria	Substantial damage to RPA
AO-2014-056	Interference from ground	22 Mar 14	Unknown remotely piloted vehicle	Near Newcastle Westpac Base (HLS), New South Wales	Nil
AO-2014-052	Interference from the ground	19 Mar 14	Unknown remotely piloted vehicle	23 km NNE Perth Airport, Western Australia	Nil
AO-2013-167	Near collision	12 Sep 13	Sensefly eBee 178	37 km SSW of Horsham aerodrome, Victoria	Nil
AO-2012-143	Airspace incursion	28 Oct 12	China Advertising Balloons Co Ltd Airship	2.7 NM E of Moorabbin Airport, Victoria	Minor damage to RPA

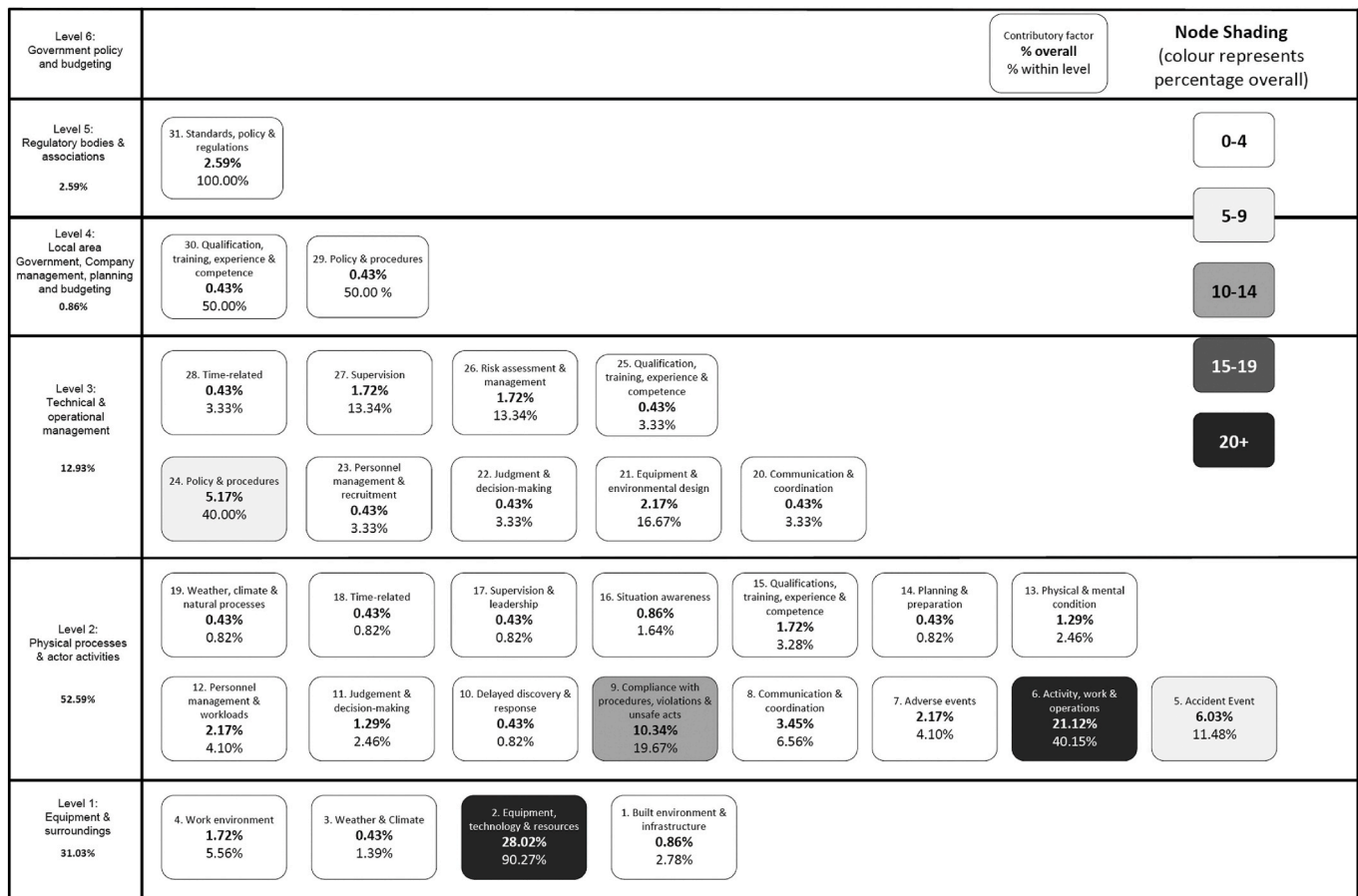


Fig. 2. Thematic contributory factors identified across the levels of the RPAS network and the proportion of factors at Level 1 (31.03%), Level 2 (52.59%), Level 3 (12.93%), Level 4 (0.86%), Level 5 (2.59%), and Level 6 (0%). Shading of nodes represents a percentage of node frequency from all nodes, where lighter colours represent lower overall frequency, and darker colours have higher overall frequency.

(overall 21.12%; within level 40.15%) and *Compliance with procedures, violations and unsafe acts* (overall 10.34%; within level 19.67%). Level 1 had the next highest proportion (31.03%), where the most frequent theme was *Equipment, technology and resources* (overall 28.02%; within level 90.27%). Level 3 had a lower overall proportion (12.93%), where the most frequent theme was *Policy and procedures* (overall 5.174%; within level 40.00%). At the higher levels of Level 4, Level 5, and Level 6, the respective proportions (0.86%, 2.59%, 0%) are also lower overall.

3.2. Network analysis of systemic contributory factors

Fig. 3 presents the aggregate AcciMap in network form, showing the relationships between all contributory factors. The connections between nodes represent the relationships between these nodes, the line's thickness and colour show the connection's strength; where the line is darker and thicker, a greater frequency of connections exists. It can be seen that the most connections between nodes are between contributory factors at Level 2.

Using outdegree and indegree centrality metrics, key nodes (those that were greater than one standard deviation above the mean) are presented in **Table 3**. For outdegree centrality, this was calculated as mean $0.032 + SD\ 0.062 = 0.094$. For indegree centrality, this was calculated as mean $0.032 + SD\ 0.066 = 0.098$. The same three contributory factor themes emerged as key nodes for both outdegree and indegree centrality measures. These key nodes are also represented in **Fig. 3** via grey shading.

3.3. Focused analysis – most commonly occurring contributory factors and relationships

Fig. 4 shows the most common contributory factors at each level. The numerical figure on the line represents the frequency of connections between contributory factors. Similarly to **Fig. 3**, the thickness of the lines indicates the frequency of relationships. Further, as noted earlier, the network analysis method also allowed for the representation of relationships between factors coded to the same theme. It can be seen that the frequency of relationships between contributory factors is generally greatest at the lowest levels of the AcciMap, compared to the upper levels. The most connected nodes at each level are as follows: *Equipment, technology and resources* (Level 1); *Activity, work and operations* (Level 2); *Policy and procedures* (Level 3); *Qualification, training, experience and competence* (Level 4); *Standards, policy and regulations* (Level 5). In examining the most frequent connections between nodes, the contributory factor *Equipment, technology and resources* (Level 1) most frequently led to other contributory factors also categorised as *Equipment, technology and resources* ($n = 41$). Similarly, the second most frequent connection was between *Activity, work and operations* (Level 2) and contributory factors also categorised as *Activity, work and operations* ($n = 20$). This was followed by the casual link from *Activity, work and operations* to *Equipment, technology and resources* ($n = 19$).

4. Discussion

This study aimed to conduct an in-depth analysis to identify the patterns of contributory factors involved in adverse RPAS events within

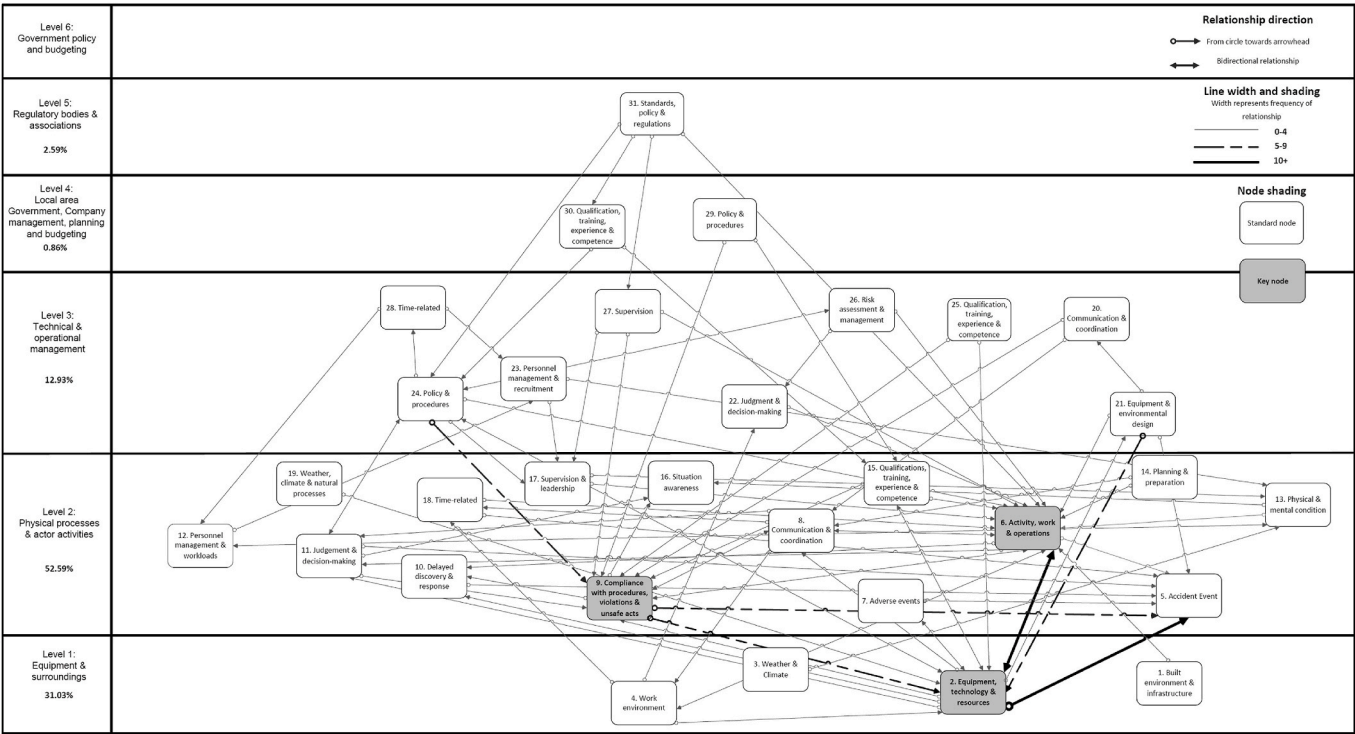


Fig. 3. Aggregate AcciMap contributory factors and relationships. The width of the links between nodes represents the frequency of causal relationships between those factors identified across the data set. The relationship direction is indicated by the arrow. If an arrow with a circle and a single arrowhead is used, it indicates the direction of the relationship (i.e., from the circle to the arrowhead). A double-ended arrow indicates a bi-directional relationship. Shaded nodes represent ‘key nodes’ for both indegree and outdegree centrality metrics.

Table 3
Key nodes were identified from outdegree centrality and indegree centrality metrics.

Level	Contributory factor theme	Value
Outdegree centrality (Degree centrality)		
Level 1. Equipment & surroundings	2. Equipment, technology & resources	0.288
Level 2. Physical processes & actor activities	6. Activity, work & operations	0.201
Level 2. Physical processes & actor activities	9. Compliance with procedures, violations & unsafe acts	0.115
Indegree centrality (Degree prestige)		
Level 1. Equipment & surroundings	2. Equipment, technology & resources	0.309
Level 2. Physical processes & actor activities	6. Activity, work & operations	0.201
Level 2. Physical processes & actor activities	9. Compliance with procedures, violations & unsafe acts	0.122

Australia. Relationships between contributory factors were examined, and key contributory factors were identified. These insights can assist in revealing systemic vulnerabilities and areas for intervention. Overall, the findings from this study support the notion that adverse events do not occur in isolation but rather from multiple decisions and actions from actors across and between system levels (Rasmussen, 1997). While this remains true, most contributory factors identified from the investigation reports were identified at the lowest levels of the RPAS.

The most frequently identified contributory factor theme across the 14 adverse RPAS events was *Equipment, technology and resources* (Level 1). The contributory factors coded to this theme differed across the investigation reports and included malfunctioning or inoperable technology, technology design (both software and hardware design), and network/communication link planning. The prominence of equipment and technology as contributing to adverse RPAS events is consistent with

previous research (Ghasri and Maghrebi, 2021; Grindley et al., 2024; Joslin, 2015; Wild et al., 2016, 2017). However, previous research used methods such as HFACS (Giese et al., 2013; Grindley et al., 2024; Manning et al., 2004; Thompson et al., 2005; Tvaryanas et al., 2006; Tvaryanas and Thompson, 2008) or other non-systems analysis methods (Feltman et al., 2018; Ghasri and Maghrebi, 2021; Joslin, 2015; Nesbit et al., 2017; Wild et al., 2016, 2017) and thus were unable to analyse the relationships between contributory factors leading to adverse RPAS incidents. This study has demonstrated the central role and tightly coupled nature of *Equipment, technology and resources* in the causal network. The network metrics highlighted *Equipment, technology and resources* as a key node in both influencing other nodes and being influenced by other nodes within system. A related notable finding is the recursive connections that *Equipment, technology and resources* has with itself. Several events in the AcciMaps illustrated how one technology issue led directly to another, demonstrating the cascading nature of technological failures. A single failure can prompt a negative spiral of events when something goes wrong (Perrow, 1999). As noted above, the technological problems emerged from different influences, including malfunctioning or inoperable technology, technology design, and network/communication links. Other influences included general operational planning (e.g. communication delay between operators due to different situational locations), environmental factors (e.g. electromagnetic interference, unstable atmospheric conditions) or compliance with procedures, violations and unsafe acts (e.g. launching RPAS with unserviceable components or being incorrectly assembled). One example from an ATSB (2022a) report involving a DJI Inspire 2 RPA illustrated this series of contributing factors beginning when the RPA compass malfunctioned, leading to other factors that relied on an accurate compass. The result was that the RPA entered an uncontrollable state, eventually impacting a hotel. In this brief sequence of events, various potential hazards related to system design are underscored. These include the lack of redundancy to control the RPA and the potential for the RPA to reach high speeds in

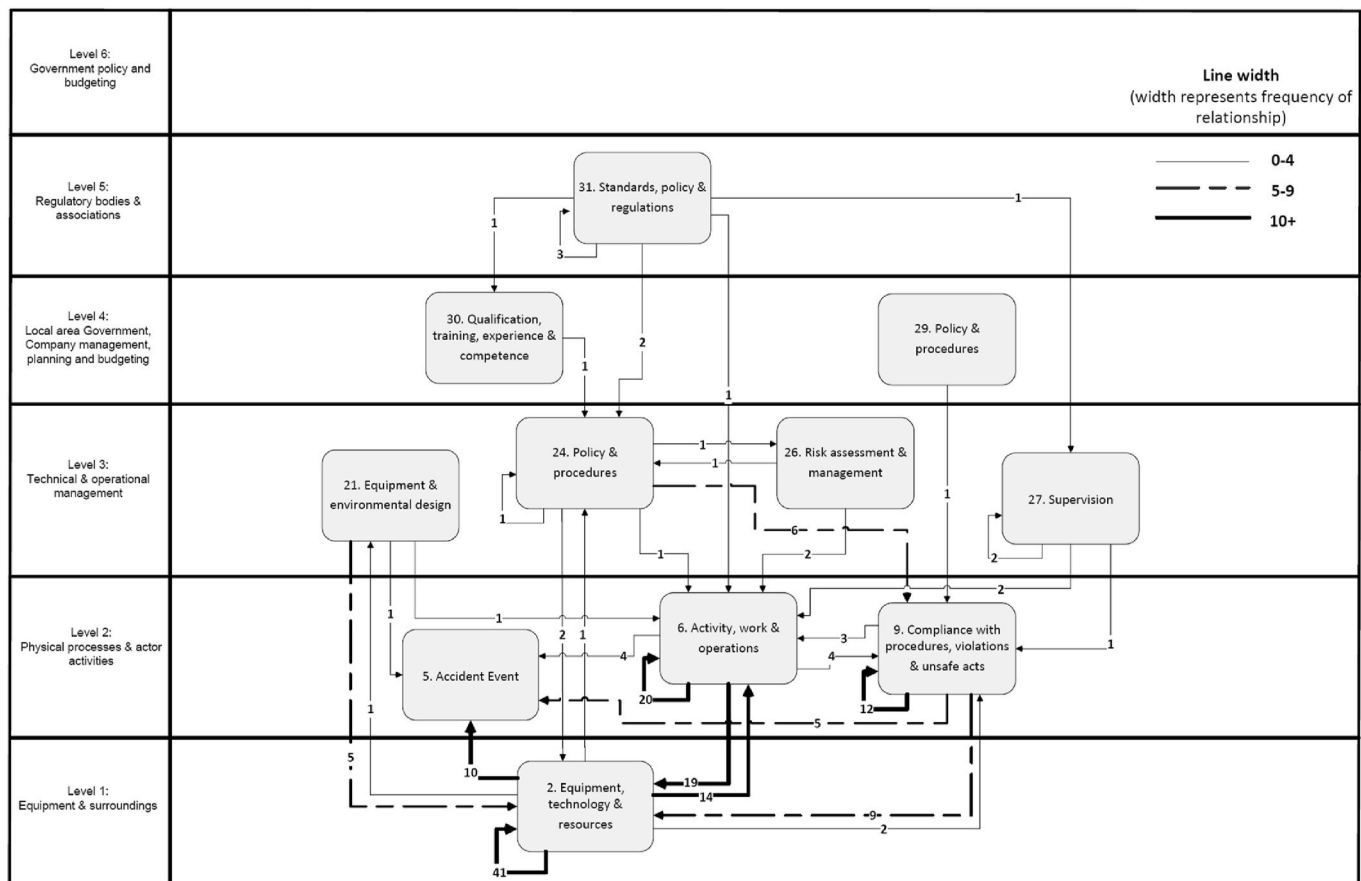


Fig. 4. Network of most common contributory factors at each level (top 10% frequency within the level; Salmon et al., 2020), including their relationships with other common nodes. The width of the links between nodes represents the frequency of causal relationships between those factors identified across the data set.

an uncontrolled state. Furthermore, these design factors may represent potential security vulnerabilities that could be exploited by malicious actors.

Many aspects of RPAS technology are relatively new and quickly developing, so it is not unreasonable to expect some challenges associated with the developing status of the technology (Albihn, 2022; Hobbs and Lyall, 2016). Minimising this risk, however, is critical to a safer system. Previous research has provided broad, non-specific suggestions regarding airworthiness controls to address equipment issues and alleviate problems (Wild et al., 2017). However, care must be taken when implementing isolated solutions involving unruly technologies like RPAS. The implementation of such suggestions may be more difficult than suggested. Current airworthiness standards and regulations are complicated and nuanced structures, and their application depends on RPAS characteristics such as size, type, system autonomy, and pace of technological development (Clothier et al., 2015). Further, RPAS technologies exist in an industry facing competitive pressures, rapid growth, and other evolving regulations (SESAR, 2024); elements known to increase risk. There remains a considerable challenge to create a robust system that accommodates safety assumptions while being flexible and adaptable to complement, rather than restrict, the technological context of an innovative and growing industry (Dekker, 2011). It should be noted that while *Equipment, technology, and resources* was the most frequent contributory factor in this sample, other systemic factors influence this outcome and should be the focus of future interventions.

Another frequently identified contributory factor theme identified in this study was *Activity, work and operations* (Level 2). The prominence of this factor shows that even normal work or actions can contribute to adverse events. It is becoming widely accepted that factors associated with the variability of normal work influences system safety (e.g. Cort,

2021; Hollnagel, 2014; Kyriakidis et al., 2018). Such decisions or actions might concurrently play an essential role in supporting system safety while still being part of a sequence of events that results in an adverse outcome. Examples of normal work in this context include conducting routine maintenance or the crew correctly identifying the state of the RPA. In these examples, the actions were precursors to another contributory factor, eventually leading to a negative outcome. Previous RPAS incident analyses using methods such as HFACS fail to address this aspect as they focus exclusively on failures. The findings, therefore, support the importance of capturing normal work performance during safety investigations rather than focusing on failure alone (Dekker, 2002a, 2002b; Hollnagel, 2014) and the use of AcciMap for accident analysis. Network metrics also reinforce the significance of *Activity, work and operations* as it is the second most influenced and influential contributory factor in the network. Again, another notable finding is the recursive connections that *Activity, work and operations* has with itself. Several events in the AcciMaps illustrated how one normal activity, work or operational factor led directly to another. Despite the routine nature of the behaviour, it constituted one part of a series of cascading events leading to an adverse outcome. An example of this, drawn from the ATSB (2022b) report involving an RF Designs Mephisto, is how a crew member directed an RPA across a runway, away from a hanger and personnel, where it eventually struck a boundary fence. In this instance, the crew's actions contributed to the adverse event but minimised the damage, which had the potential to be much more significant.

A third frequently identified contributory factor theme was *Compliance with procedures, violations, and unsafe acts* (Level 2). Factors in this theme were broad but included errors (e.g. missing technical safety indicators during set-up) and non-compliance with policy, procedures, and operating limitations (e.g. mishandling during transport). In the

network of contributory factors identified, another recursive connection is shown from *Compliance with procedures, violations and unsafe acts* (Level 2) to itself. An example of this can be seen in the [ATSB \(2022a\)](#) report involving the DJI Inspire 2, where the pilot did not try to use the Smart Return to Home function or emergency engine shutdown procedure. This led to the recursive connection that the pilot did not switch the RPA to automatic mode, which the manufacturer advised would have allowed them to regain control of the RPA. The RPA then impacted a hotel. This series of events potentially shows several missed opportunities to regain control of the RPA before it impacted the hotel, but care must be taken in attributing blame, as this must still be considered in the context of the entire scenario, hindsight bias, and local rationality (i.e. What made sense to the pilot at the time; [Dekker, 2002a](#)). A series of factors such as this should raise questions about higher-level influencing factors such as culture, training, or external systems pressures not always examined in investigations. Previous research has linked factors such as the quality of training programs to confidence levels and flying performance of operators, contributing to accidents ([Li et al., 2022](#)). In this study, the method of aggregating AcciMap data uncovers some of these relationships. For example, the relationships between *Policy and procedures* (Level 3) and *Compliance with procedures, violations and unsafe acts* (Level 2).

The outward causal relationship from *Compliance with procedures, violations and unsafe acts* (Level 2) to *Equipment, technology and resources* (Level 1) highlights the increased potential for cascading negative events if something goes wrong with the equipment, as discussed in the recursive connections between *Equipment, technology and resources*. An example of this casual connection is drawn from the [ATSB \(2019\)](#) report involving an Insitu ScanEagle X200. In this instance, the RPA was launched with an unserviceable pitot-static system, which led to the equipment contributory factor where *Unreliable airspeed data led to the RPA entering an aerodynamic stall*. Despite efforts from the crew to recover the aircraft, a series of other *Equipment, technology and resources* contributory factors led to the RPA colliding with the ground. The inward causal relationship from *Policy and procedures* (Level 3) towards *Compliance with procedures, violations and unsafe acts* (Level 2) demonstrates the potential impact of decisions and actions of all stakeholders on systems outcomes. An example of this is drawn from the [ATSB \(2017b\)](#) report involving the Pulse Aerospace Vapor 55, where the Default hemisphere was set to North for entering positions in the Ground Control Station, but this information was not included in the RPA flight manual and thus unknown to the pilot. This led directly to an error occurring where *the South-eastern point used to geo-reference the image on the GCS map was incorrectly assigned to a northern hemisphere latitude, which resulted in incorrect waypoints and home position for the mission*. The final result was that the RPA was lost. Previous research has also identified latent system failures at these higher levels, such as policies, procedures, and design specifications. In some instances, these shortfalls also lead to RPAS equipment, technology, and resource breakdowns ([Tvaryanas et al., 2006](#)). This research shows the important role of *Compliance with procedures, violations and unsafe acts* (Level 2) as an intermediary factor in adverse outcomes. The identification of this causal pathway, with its high degree of connectedness, may present a powerful opportunity to improve system safety ([Salmon et al., 2020](#)).

4.1. Systemic vulnerabilities leading to adverse RPAS events

As noted above, while some higher-level contributory factors were identified from the data analysed, they were few. Further, no contributory factors were identified at Level 6 in any reports. This contrasts systems theory, such as that outlined in the RMF ([Rasmussen, 1997](#)), where we expect to see contributing factors at all system levels, including the higher levels. Research by [Salmon et al. \(2020\)](#), showing numerous contributory factors at all system levels, also suggests there are likely to be more higher-level factors than what was identified in this study. Fewer contributory factors identified at the higher levels are

likely exacerbated by the challenges of conducting field investigations, where resources like time and the scope of investigations are often limited. As a result, higher system levels may not be fully investigated, especially in adverse events that do not involve significant financial or human loss ([Grindley et al., 2024](#)). Although fewer contributory factors were identified at the higher levels in this study, there were still important causal paths across five of the six levels highlighted using the AcciMap method. This contrasts with traditional methods such as HFACS, which uses a taxonomy to consider factors such as unsafe acts, training, experience and competence of human operators with limited focus on the role of stakeholders at higher levels ([Hulme et al., 2019](#); [Salmon et al., 2020](#)). For example, previous research using HFACS to analyse RPAS incidents identified the most significant contributing factors existing at the 'pointy end'. These were operator errors, including skill-based errors and decision errors, followed by the technical environment ([Giese et al., 2013](#); [Grindley et al., 2024](#)). Other cognitive and psycho-social factors affecting performance were also highlighted as significant contributing factors, such as attention, communication and overconfidence ([Thompson et al., 2005](#); [Tvaryanas et al., 2006](#)). Although important higher-level contributory factors were identified in some of the studies, such as resource and acquisition management policies and processes ([Thompson et al., 2005](#); [Tvaryanas et al., 2006](#)), regulatory factors and organisational climate ([Grindley et al., 2024](#)), there was a disconnect in the causal link between these higher-level contributory factors and the actions/outcomes at lower levels ([Grindley et al., 2024](#); [Thompson et al., 2005](#); [Tvaryanas et al., 2006](#)).

The previous aggregate AcciMap analyses conducted by [Salmon et al. \(2020\)](#) and [Read et al. \(2022\)](#) both found that most contributory factors and connections were identified at the lowest levels. However, both of these previous studies identified many more contributory factors at higher levels than was found here. Examples of possible higher-level contributory factors not identified in this study included *Audits and inspections* (Level 5) and *Budget and finance* (Level 6). Unsurprisingly, [Salmon et al. \(2020\)](#) found numerous factors at the higher levels, as the analysis undertaken to develop the thematic classification scheme was based on previously published AcciMaps, specifically designed to identify factors across system. A notable similarity in key contributory factors between [Read et al. \(2022\)](#), [Salmon et al. \(2020\)](#), and the current study is the prominence of the contributory factor *Compliance with procedures, violations, and unsafe acts* (Level 2). Further, this is a key node for both indegree and outdegree centrality in all three papers, emphasising this overarching theme's key role in influencing and being influenced by other contributory factors leading to adverse events. This may simply reflect the practical ease of identifying the decisions and actions of the 'last person to touch the system' ([Read et al., 2021](#)), rather than telling us anything profound about accident causation. There are many possible systemic influences leading to these behaviours in the RPAS context. These may include the level of system maturity and the fast pace of development that have led to inadequate training programs, policies and procedures, and regulations and standards. While previous research has not necessarily identified such factors as contributory factors, system changes in these areas, such as improving technical publications, checklists, and initial operator training programs, have been recommended ([Tvaryanas et al., 2006](#)).

4.2. Safety improvement

While there are notable vulnerabilities in the current RPAS system highlighted above, opportunities to improve safety are also found. By mapping the RPAS sociotechnical system with AcciMap its tightly coupled and complex nature can be seen. In such systems, vertical integration, including effective feedback mechanisms, across levels of the RPAS sociotechnical system is paramount ([Rasmussen, 1997](#); [Viceinte and Christoffersen, 2006](#)). The causal network of contributory factors shown in this study hints at a lack of feedback and vertical

integration across all levels of the RPAS. Further analysis and changes to the feedback mechanisms across the RPAS system may be warranted to enhance vertical integration. Currently, it may be difficult for actors at each level to see how their decisions interact with those made by actors at other levels. In this case, the threats to safety may be obscure or ambiguous before an adverse event (e.g., regulatory authorities pass new requirements that do not integrate feedback from companies; thus, requirements are not implemented as intended at the operational level). A coherent, vertically integrated structure (Rasmussen, 1997) that is transparent and comprehensible must be in place to reduce risk and enable effective system functioning and development. However, caution must be exercised when implementing individual risk controls, which could result in the system becoming a complex patchwork of temporary solutions that fail to consider the intricate interactions within the system, particularly those that may involve a time delay (Sterman, 2000). Building a complex system or integrating it into an existing system requires careful fine-tuning and constant measuring of controls and feedback loops to achieve longer-term improvement goals (Meadows, 2008; Sterman, 2000). Ideally, a coordinated approach to the design of system-wide risk controls should be undertaken, covering all aspects, including legislation, regulations, research and development, and policies/procedures, before any specific recommendations for system improvement can be made.

The safe and effective integration of the RPAS sector into national airspace has been a considerable challenge for states around the globe (EUROCONTROL, 2018; FAA, n.d.) which have applied different approaches. RPAS regulation and safety systems development in Australia and internationally are still in the early phases (CASA, 2022; Clothier et al., 2015). At present, there is a high rate of change in the current system as it evolves. Relevant Australian government agencies have taken active steps to integrate the rapidly evolving aviation environment to better include traditional aviation and remote-piloted aviation (Airservices Australia, 2022; CASA, 2022; Department of Infrastructure, Transport, Regional Development and Communications, 2021). Findings like those presented in this paper can inform the discussion surrounding the appropriate structure for system safety that promotes vertical integration, as discussed above. Previous studies have highlighted the importance of higher system-level safety controls, such as legislation (Parnell et al., 2017), regulation (Blanc et al., 2022), guidelines and standards (Hobbs and Lyall, 2016; Hobbs and Shively, 2013), and other aspects of the safety management system (Stolzer et al., 2011). Nevertheless, these controls can only be effective protective measures when they are reputable and relevant (i.e. up-to-date and suitable to address the specific problem); outdated or unreliable controls can create further system problems (Dekker, 2011), emphasising the need for high-quality feedback systems and ongoing system improvement. The research community can contribute to addressing this problem by conducting further systems analyses to expose the current system state before making specific recommendations to change the system. Ongoing analysis with the AcciMap technique can continue to build knowledge about contributory factors to adverse RPAS events as the system expands and evolves. Additionally, systems-based methods such as the Systems-Theoretic Accident Model and Processes (STAMP; Leveson, 2004) and Systems Theoretic Process Analysis (STPA; Leveson, 2011) offer the capacity to model complex and dynamic systems prospectively. Future research could model Australia's existing RPAS safety control structure to identify current controls and feedback mechanisms. Interventions can then focus on harmonising system pressures of balancing safety obligations with pushing the boundaries in innovation, research and development. Future research of this nature will extend the effectiveness of the research question posed in this article, exposing more of the vulnerabilities and risks within the current Australian RPAS safety system.

We strongly believe full systems analyses are required before specific recommendations for change are made. Nevertheless, we acknowledge the emerging risks associated with RPAS operation and the need to make

gradual changes to reduce risks and adapt alongside the technology. To that end, we propose some general recommendations for stakeholders to consider at this stage based on the findings of this research. We support the notions highlighted earlier by other researchers that there should be a focus on developing higher-level safety controls as a starting point (Blanc et al., 2022; Hobbs and Lyall, 2016; Hobbs and Shively, 2013; Parnell et al., 2017; Stolzer et al., 2011). To reduce incidences of equipment and technology failures, regulators should establish standards and regulations to ensure robust design, testing, and maintenance practices are followed (Clothier et al., 2015). They must also ensure that manufacturers, companies and operators have appropriate facilities, such as *sandboxes*, to test and validate emerging technologies (CAA, 2023; CASA, 2022). The significance of data capture as a feedback mechanism to inform decision-making is essential. To that end, reporting and monitoring systems should be implemented at all levels to provide adequate feedback to policymakers, legislators and other decision-makers (ICAO, 2015; Wild et al., 2017) – this may be legislated as a requirement, or it may be a proactive implementation by companies and manufacturers as a part of their safety management system – regardless, it is key to effective decision-making at all levels. A lack of thorough investigation and reporting on higher system-level factors results in the failure to identify systemic issues and implement effective interventions, leading to missed opportunities to mitigate risks and enhance system safety and performance. Investigations and incident reporting are critical feedback mechanisms to measure system performance (ICAO, 2013), especially when the data can be aggregated and trends identified. In Australia, government-level investigative bodies, like the ATSB, play a key role in this data collection and analysis. However, responsibility for this critical feedback mechanism should be shared. If researchers develop reliable, valid and easy-to-use analysis methods, then practitioners, manufacturers and companies can implement these internally to look deeper at the problems (Grindley et al., 2024). Additionally, educating stakeholders and investigators on systems thinking and accident causation is likely to enhance the consideration of higher-level factors (Salmon et al., 2017). Improved collaboration, transparency and information sharing between stakeholders may lead to significant insights. Data collection of higher-level contributory factors might also be improved through interdisciplinary investigation teams (with members internal and external to organisations; ESReDA, n.d.), mandated safety reporting measures (ICAO, 2013), and enhanced networking and collaborative data management techniques between multiple organisations (Osório et al., 2022). Ensuring effective data collection requires proper funding from governments and company management to support bodies like the ATSB and facilitate internal company investigations, promoting a continuous learning and improvement process. These general recommendations are just some drawn from the findings highlighted in this study to improve system safety and performance. Further higher-level systems analyses are likely to reveal other, more targeted recommendations.

4.3. Study limitations

A limitation of this study was the small sample size ($n = 14$) of adverse events, as only full reports published by the ATSB were included. While including reported events that were not fully investigated may provide a broader picture of contributing factors, the drawback of using such reports is the difficulty in identifying the network of latent contributory factors due to the lack of detailed investigation findings. Including full investigation reports from other international investigatory bodies might have improved the fidelity of the findings but would need to address any country-specific differences in RPAS operations and the policy and regulatory context surrounding their use. Future studies may consider taking an international focus and potentially comparing the contributory factors to RPAS incidents identified in reports by organisations such as the UK Air Accidents Investigation Branch (AAIB), the US National Transportation Safety Board (NTSB), and the

French Bureau d'Enquêtes et d'Analyses (BEA).

A second limitation relates to the reliance on investigation reports as a data source. While the ATSB is recognised for producing high-quality reports (ATSB, 2008; Underwood and Waterson, 2014), the resources of investigation bodies are limited. Therefore, the scope of investigations may restrict the number of contributory factors identified or reported at higher system levels. In particular, within the sample of reports analysed in this paper, higher-level contributing factors associated with government-level decisions and actions were not identified. Potentially, valuable insight would be gained from looking at these higher levels in more depth in future investigations (Grindley et al., 2024). Furthermore, additional data sources suitable for in-depth adverse event analysis could include internal company investigation reports. Collaborating with stakeholders might enhance data collection and provide insights into higher-level system contributory factors. Future research could also explore contributory factors at the organisational, regulatory, and government levels of RPAS in Australia via engagement with subject matter experts operating at these system levels.

5. Conclusion

This study identified a network of contributory factors leading to adverse RPAS events in Australia through a novel application of the aggregate AcciMap method. It supports past research that highlights specific issues within this complex system. The analysis also extends the literature by illustrating the causal network linking contributory factors between higher and lower levels of the RPAS system. However, further work is required to identify more higher-level contributory factors. The

findings highlight system vulnerabilities and opportunities to improve safety. In particular, opportunities for improvement include establishing and monitoring more effective safety structures to enhance vertical integration of the system, providing the potential to reduce risk and optimise performance outcomes. Careful consideration and continuous monitoring of the system's control and feedback mechanisms are necessary to ensure a safe yet practical protective system within a complex and dynamic industry.

CRediT authorship contribution statement

Samantha L. Jackson: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Paul M. Salmon:** Writing – review & editing, Supervision. **Gemma J.M. Read:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The Commonwealth of Australia funded Samantha Jackson's contribution to this research through the Australian Government Research Training Program Scholarship.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apergo.2025.104468>.

Appendix A

Thematic contributory factor types for aggregate AcciMap (from Salmon et al., 2020).

Level 1: Equipment & surroundings Animal, plant & biological hazards Built environment & infrastructure Equipment, technology & resources Information & data Noise & visibility Other Physical & natural environment Time-related Weather & climate Work environment	Level 4: Local area government, company management, planning and budgeting Communication & coordination Compliance with procedures, violations & unsafe acts Culture Financial pressures Judgement & decision-making Other Personnel management & recruitment Planning & preparation Policy & procedures Qualification, training, experience & competence Risk assessment & management Supervision Time-related
Level 2: Physical processes & actor activities Accident event Activity, work & operations Adverse events Communication & coordination Compliance with procedures, violations & unsafe acts Delayed discovery & response Equipment, technology & environment Group & teamwork Judgement & decision-making Other Personal management & workloads Physical & mental condition Planning & preparation Qualification, training, experience & competence Risk assessment & management Situation awareness Supervision & leadership	Level 5: Regulatory bodies & associations Audits & inspections Communication & coordination Compliance with procedures, violations & unsafe acts Culture Financial pressures Judgement & decision-making Planning & preparation Qualification, training, experience & competence Regulatory structures & services Risk assessment & management Standards, policy & regulations Time-related Unclear roles & responsibilities

(continued on next page)

(continued)

Time-related
Weather, climate & natural processes

Level 3: Technical and operational management

Communication & coordination
Compliance with procedures, violations & unsafe acts
Culture
Equipment & environment design
Financial pressures
Judgement & decision making
Other
Personnel management & recruitment
Planning & preparation
Policy & procedures
Qualification, training, experience & competence
Risk assessment & management
Supervision
Time-related

Level 6: Government policy & budgeting

Action omitted & failure to act
Budget & finance
Communication & coordination
Culture
Judgement & decision-making
Policy, legislation & regulation
Political structures & services
Priorities
Qualification, training, experience & competence
Supervision & enforcement

References

- Airservices Australia, 2022. Connecting people with their world safely. <https://www.airservicesaustralia.com/what-we-do/>.
- Aibih, Joakim, 2022. Human factors in aerial drone operations [Lund University]. <http://lup.lub.lu.se/student-papers/search/publication/9094702>.
- ATSB. (n.d.). Occurrence investigation levels [Australian Transport Safety Bureau official website]. <https://www.atsb.gov.au/occurrence-investigation-levels>.
- ATSB, 2008. *Analysis, Causality And Proof In Safety Investigations* (Aviation Research and Analysis Report No. AR-2007-053; ATSB Transport Safety Research Report). Australian Transport Safety Bureau. <https://www.atsb.gov.au/sites/default/files/media/27767/ar2007053.pdf>.
- ATSB, 2013. Airspace incursion involving unmanned airship, Airship 11 (Aviation Occurrence Investigation No. AO-2012-143). Australian Transport Safety Bureau. https://www.atsb.gov.au/publications/investigation_reports/2012/air/ao-2012-143.
- ATSB, 2014a. Aircraft Separation Issue Involving an Ayres S2R, VH-WBK and an Unmanned Aerial Vehicle (Aviation Occurrence Investigation No. AO-2013-167, 7). Australian Transport Safety Bureau. https://www.atsb.gov.au/publications/investigation_reports/2013/air/ao-2013-167.
- ATSB, 2014b. Near collision involving an unmanned aerial vehicle and a Bell 412. VH-WSR (Aviation Occurrence Investigation No. AO-2014-056). Australian Transport Safety Bureau. https://www.atsb.gov.au/publications/investigation_reports/2014/air/ao-2014-056.
- ATSB, 2017a. A safety analysis of remotely piloted aircraft systems (No. AR-2017-016a). Australian Trans. Safe. Bureau. <https://www.atsb.gov.au/publications/2017/ar-2017-016>.
- ATSB, 2017b. Loss of control involving remotely piloted aircraft Pulse Aerospace vapor 55 (aviation occurrence investigation No. AO-2016-128). Australian Trans. Safe. Bureau. https://www.atsb.gov.au/publications/investigation_reports/2016/air/ao-2016-128.
- ATSB, 2019. Collision with terrain involving an Insitu ScanEagle X200 unmanned aircraft system (UAS). Aviation Occurrence Investigation No. AO-2019-004). Australian Transport Safety Bureau. https://www.atsb.gov.au/publications/investigation_reports/2019/air/ao-2019-004.
- ATSB, 2020. In-flight break-up involving Airbus Zephyr unmanned aerial vehicle. Aviation Occurrence Investigation. (Short) No. AO-2019-056). https://www.atsb.gov.au/media/5778702/ao-2019-056_final.pdf.
- ATSB, 2022a. Loss of control and collision with terrain involving DJI Inspire 2 remotely piloted aircraft (Aviation Occurrence Investigation (Defined) No. AO-2021-001). Australian Trans. Safe. Bureau. https://www.atsb.gov.au/publications/investigation_reports/2021/air/ao-2021-001.
- ATSB, 2022b. Loss of control during taxi, involving RF Designs Mephisto remotely piloted aircraft (Aviation Occurrence Investigation (Defined) No. AO-2020-035). Australian Trans. Safe. Bureau 26. https://www.atsb.gov.au/publications/investigation_reports/2020/air/ao-2020-035.
- ATSB, 2023. ATSB Annual Report 2022-23. Australian Transport Safety Bureau.
- Blanc, Florentin, Ottimofiore, Giuseppa, Myers, Kevin, 2022. From OSH regulation to safety results: using behavioral insights and a “supply chain” approach to improve outcomes – the experience of the health and safety Executive. Saf. Sci. 145, 105491. <https://doi.org/10.1016/j.ssci.2021.105491>.
- CAA. Regulatory sandbox for the development of capabilities to integrate Unmanned Aerial Systems (UAS) in unsegregated airspace. UK Civil Aviation Authority. <https://www.caa.co.uk/publication/download/20949>.
- CASA, 2022. RPAS and AAM Strategic Regulatory Roadmap—Public Consultation.
- Cho, George, 2013. Unmanned aerial vehicles: emerging policy and regulatory issues. J. Law Inf. Sci. 22 (2), 201–236.
- Cilliers, Paul, 1998. Complexity and Postmodernism Understanding Complex Systems. Routledge.
- Clarke, Roger, Bennett, Moses, Lyria, 2014. The regulation of civilian drones’ impacts on public safety. Comput. Law Secur. Rep. 30 (3), 263–285. <https://doi.org/10.1016/j.clsr.2014.03.007>.
- Clothier, Reece A., Williams, Brendan P., Coyne, James, Wade, Mark, Washington, Achim, 2015. Challenges to the development of an airworthiness regulatory framework for unmanned aircraft systems. 16th Australian Aerospace Congress, Melbourne.
- Clothier, Reece A., Williams, Brendan P., Hayhurst, Kelly J., 2018. Modelling the risks remotely piloted aircraft pose to people on the ground. Saf. Sci. 101, 33–47. <https://doi.org/10.1016/j.ssci.2017.08.008>.
- Cooke, Nancy J., Rowe, Leah J., Bennett, Winston, Joralmon, DeForest Q. (Eds.), 2017. Remotely Piloted Aircraft Systems: A Human Systems Integration Perspective. Wiley.
- Cort, Rebecca, 2021. Getting work done: the significance of the human in complex socio-technical systems. Digital Comprehensive Summaries of Uppsala Dissertations from the Faculty of Science and Technology 2088. Uppsala: Acta Universitatis Upsaliensis. ISBN 978-91-513-1330-6.
- Dekker, Sidney, 2002a. Reconstructing human contributions to accidents: the new view on error and performance. J. Saf. Res. 33 (3), 371–385. [https://doi.org/10.1016/S0022-4375\(02\)00032-4](https://doi.org/10.1016/S0022-4375(02)00032-4).
- Dekker, Sidney, 2002b. The Field Guide to Human Error Investigations. Ashgate.
- Dekker, Sidney, 2011. Drift into Failure: from Hunting Broken Components to Understanding Complex Systems. Ashgate Pub.
- Department of Infrastructure, Transport, Regional Development and Communications, 2021. National Strategic Airspace. National Aviation Policy Issues Paper. <https://www.infrastructure.gov.au/sites/default/files/documents/national-strategic-airspace-national-aviation-policy-issues-paper.pdf>.
- EASA, 2021. Drone Incident Management at Aerodromes. Part 1: The Challenge of Unauthorised Drones in the Surroundings of Aerodromes (No. Ref. Ares(2022)3535298-09/05/2022). European Union Aviation Safety Agency. <https://www.easa.europa.eu/en/drone-incident-management-aerodromes-part-1>.
- ESReDA. (n.d.). Guidelines for safety investigations of accidents. European Safety Reliability and Data Association.
- EUROCONTROL, 2018. UAS ATM integration. Operational Concept, 1.0.
- FAA. FAA Aerospace Forecast. Fiscal Years 2021–2041. https://www.faa.gov/data_research/aviation/aerospace_forecasts/media/FY2021-41_FAA_Aerospace_Forecast.pdf.
- Feltman, Kathryn A., Curry, Ian, Kelley, Amanda M., 2018. Review of U.S. Army unmanned aerial systems accident reports: Analysis of human error contributions (No. USAARL report No. 2018-08). United States Army Aeromedical Research Laboratory. <https://apps.dtic.mil/sti/trecms/pdf/AD1052734.pdf>.
- Fortune Business Insights, 2023. Unmanned Aerial Vehicle (UAV) Market Research Report. No. FBI01603).
- Ghasri, Milad, Maghrebi, Mojtaba, 2021. Factors affecting unmanned aerial vehicles’ safety: a post-occurrence exploratory data analysis of drones’ accidents and incidents in Australia. Saf. Sci. 139, 105273. <https://doi.org/10.1016/j.ssci.2021.105273>.
- Giese, Stefanie, Carr, David, Chahl, Javaan, 2013. Implications for unmanned systems research of military UAV mishap statistics. IEEE Intelligent Vehicles Symposium (IV), 1191–1196. <https://doi.org/10.1109/IVS.2013.6629628>, 2013.
- Grindley, Ben, Phillips, Katie, Parnell, Katie J., Cherrett, Tom, Scanlan, Jim P., Plant, Katherine L., 2024. A decade of uav incidents: a human factors analysis of casual factors. Appl. Ergon. 121. <https://doi.org/10.1016/j.apergo.2024.104355>.
- Hanneman, Robert A., Riddle, Mark, 2005. Introduction to Social Network Methods. University of California, Riverside. <http://faculty.ucr.edu/~hanneman/>.
- Hobbs, Alan, 2010. Unmanned aircraft systems. In: Human Factors in Aviation, second ed. Academic Press, pp. 505–531.
- Hobbs, Alan, Lyall, Beth, 2016. Human Factors Guidelines for Remotely Piloted Aircraft System (RPAS) Remote Pilot Stations (RPS) (No. Guidelines. San Jose State University Research Foundation/NASA Ames Research Centre, version 1.1. <https://ntrs.nasa.gov/api/citations/20190000211/downloads/20190000211.pdf>.
- Hobbs, Alan, Shively, R. Jay, 2013. Human factors guidelines for UAS in the national airspace system. AUVSI Unmanned Systems 1, 567–575.

- Hollnagel, Erik, 2014. *Safety-I and Safety-II: the Past and Future of Safety Management*. Ashgate Publishing Company.
- Houghton, Robert J., Baber, Chris, McMaster, Richard, Stanton, Neville A., Salmon, Paul, Stewart, Rebecca, Walker, Guy, 2006. Command and control in emergency services operations: a social network analysis. *Ergonomics* 49 (12–13), 1204–1225. <https://doi.org/10.1080/00140130600619528>.
- Hulme, Adam, Stanton, Neville A., Walker, Guy H., Waterson, Patrick, Salmon, Paul M., 2019. Accident analysis in practice: a review of Human Factors Analysis and Classification System (HFACS) applications in the peer reviewed academic literature. *Proc. Hum. Factors Ergon. Soc. Annu. Meet.* 63 (1), 1849–1853. <https://doi.org/10.1177/1071181319631086>.
- Hulme, Adam, Stanton, Neville A., Walker, Guy H., Waterson, Patrick, Salmon, Paul M., 2024. Testing the reliability of accident analysis methods: a comparison of AcciMap, STAMP-CAST and AcciNet. *Ergonomics* 67 (5), 695–715. <https://doi.org/10.1080/00140139.2023.2240048>.
- ICAO, 2013. *Safety management manual (SMM)* (3. International Civil Aviation Organization).
- ICAO, 2015. *Manual On Remotely Piloted Aircraft Systems (RPAS)* (No. Doc 10019 AN/507). International Civil Aviation Organization.
- Johnson, Jack A., Svach, Megan R., Brown, Lawrence H., 2019. Drone and other hobbyist aircraft injuries seen in U.S. Emergency departments, 2010–2017. *Am. J. Prev. Med.* 57 (6), 826–829. <https://doi.org/10.1016/j.amepre.2019.06.023>.
- Joslin, Robert, 2015. Synthesis of unmanned aircraft systems safety reports. *J. Aviation Technol. Eng.* 5 (1), 2. <https://doi.org/10.7771/2159-6670.1117>.
- Knoke, D., Burt, R.S., 1983. Prominence. In: Burt, R.S., Minor, M.J. (Eds.), *Applied Network Analysis: A Methodological Introduction*. Sage.
- Kyriakidis, Miltos, Kant, Vivek, Amir, Sulfikar, Dang, Vinh N., 2018. Understanding human performance in sociotechnical systems – steps towards a generic framework. *Saf. Sci.* 107, 202–215. <https://doi.org/10.1016/j.ssci.2017.07.008>.
- Lenné, Michael G., Salmon, Paul M., Liu, Charles C., Trotter, Margaret, 2012. A systems approach to accident causation in mining: an application of the HFACS method. *Accid. Anal. Prev.* 48, 111–117. <https://doi.org/10.1016/j.aap.2011.05.026>.
- Leveson, Nancy, 2004. A new accident model for engineering safer systems. *Saf. Sci.* 42 (4), 237–270. [https://doi.org/10.1016/S0925-7535\(03\)00047-X](https://doi.org/10.1016/S0925-7535(03)00047-X).
- Leveson, Nancy, 2011. *Engineering a Safer World: Systems Thinking Applied to Safety*. MIT Press.
- Li, Songpo, Cummings, Mary L., Welton, Benjamin, 2022. Assessing the impact of autonomy and overconfidence in UAV first-person view training. *Appl. Ergon.* 98, 103580. <https://doi.org/10.1016/j.apergo.2021.103580>.
- Manning, Sharon D., Rash, Clarence E., LeDuc, Patricia A., Noback, Robert K., McKeon, Joseph, 2004. *The Role of Human Causal Factors in U.S. Army Unmanned Aerial Vehicle Accidents* (USAARL Report Nos. 2004–11. U.S. Army Aeromedical Research Laboratory, p. 30.
- Meadows, Donella H., 2008. Thinking in systems: a primer. In: Wright, Diana (Ed.), *Chelsea Green Pub.*
- Nesbit, Paul R., Barchyn, Thomas E., Hugenholtz, Chris H., Cripps, Sterling, Kucharczyk, Maja, 2017. Reported UAV incidents in Canada: analysis and potential solutions. *J. Unmanned Veh. Syst.* 5 (2), 51–61. <https://doi.org/10.1139/juvs-2016-0033>.
- Neville, Kelly, Williams, Kevin, 2016. Integrating remotely piloted aircraft systems into the national airspace system. In: *Remotely Piloted Aircraft Systems: A Human Systems Integration Perspective*.
- Osório, A. Luís, Antunes, Cláudia, Camarinha-Matos, Luís M., Gonçalves, Carlos, 2022. Collaborative management of traffic accidents data for social impact analytics. In: Camarinha-Matos, Luís M., Ortiz, Angel, Boucher, Xavier, Luís Osório, A. (Eds.), *Collaborative Networks in Digitalization and Society 5.0*, vol. 662. Springer International Publishing, pp. 230–241. https://doi.org/10.1007/978-3-031-14844-6_19.
- Ottino, J.M., 2003. Complex systems. *AIChE J.* 49 (2), 292–299. <https://doi.org/10.1002/aic.690490202>.
- Parnell, Katie J., Stanton, Neville A., Plant, Katherine L., 2017. What's the law got to do with it? Legislation regarding in-vehicle technology use and its impact on driver distraction. *Accid. Anal. Prev.* 100, 1–14. <https://doi.org/10.1016/j.aap.2016.12.015>.
- Perrow, Charles, 1999. *Normal Accidents: Living With High-Risk Technologies ; with a New Afterword and a Postscript on the Y2K Problem* (Repr.). Princeton Univ. Press.
- Pyrghies, John, 2019. The UAVs threat to airport security: risk analysis and mitigation. *J. Airl. Airt. Manag.* 9 (2), 63. <https://doi.org/10.3926/jairm.127>.
- Qureshi, Zahid H., Campbell, Alistair, 2009. 1.1.3 systemic safety and accident modelling of complex socio-technical systems. *INCOSE Internat. Symp.* 19 (1), 21–35. <https://doi.org/10.1002/j.2334-5837.2009.tb00935.x>.
- Rasmussen, Jens, 1997. Risk management in a dynamic society: a modelling problem. *Saf. Sci.* 27 (2–3), 183–213. [https://doi.org/10.1016/S0925-7535\(97\)00052-0](https://doi.org/10.1016/S0925-7535(97)00052-0).
- Read, Gemma J.M., O'Brien, Alison, Stanton, Neville A., Salmon, Paul M., 2022. Learning lessons for automated vehicle design: using systems thinking to analyse and compare automation-related accidents across transport domains. *Saf. Sci.* 153, 105822. <https://doi.org/10.1016/j.ssci.2022.105822>.
- Read, Gemma J.M., Shorrock, Steven, Walker, Guy H., Salmon, Paul M., 2021. State of science: evolving perspectives on 'human error'. *Ergonomics* 64 (9), 1091–1114. <https://doi.org/10.1080/00140139.2021.1953615>.
- Read, Gemma J.M., Stevens, Erin L., Lenné, Michael G., Stanton, Neville A., Walker, Guy H., Salmon, Paul M., 2018. Walking the talk: comparing pedestrian 'activity as imagined' with 'activity as done'. *Accid. Anal. Prev.* 113, 74–84. <https://doi.org/10.1016/j.aap.2018.01.016>.
- Salmon, Paul M., Cornelissen, Miranda, Trotter, Margaret J., 2012. Systems-based accident analysis methods: a comparison of Accimap, HFACS, and STAMP. *Saf. Sci.* 50 (4), 1158–1170. <https://doi.org/10.1016/j.ssci.2011.11.009>.
- Salmon, Paul M., Goode, Natassia, Taylor, Natalie, Lenné, Michael G., Dallat, Clare E., Finch, Caroline F., 2017. Rasmussen's legacy in the great outdoors: a new incident reporting and learning system for led outdoor activities. *Appl. Ergon.* 59, 637–648. <https://doi.org/10.1016/j.apergo.2015.07.017>.
- Salmon, Paul M., Hulme, Adam, Walker, Guy H., Waterson, Patrick, Berber, Elise, Stanton, Neville A., 2020. The big picture on accident causation: a review, synthesis and meta-analysis of AcciMap studies. *Saf. Sci.* 126, 104650. <https://doi.org/10.1016/j.ssci.2020.104650>.
- Salmon, Paul M., Stanton, Neville A., Walker, Guy H., Hulme, Adam, Goode, Natassia, Thompson, Jason, Read, Gemma J.M., 2022. *Handbook of Systems Thinking Methods*. CRC PRESS. <https://doi.org/10.1201/9780429281624>.
- Scyne Advisory, 2024. Sizing the future drone and advanced air mobility market in Australia. <https://www.airservicesaustralia.com/wp-content/uploads/2024/02/Sizing-the-Future-Drone-Industry-in-Australia-February.pdf>.
- SESAR, 2024. European ATM master plan. SESAR Joint Undertaking. <https://www.sesarj.eu/MP2024proposal>.
- Shackle, Samira, 2020. The mystery of the Gatwick drone. *The Guardian Newspaper*. <https://www.theguardian.com/uk-news/2020/dec/01/the-mystery-of-the-gatwick-drone>.
- Shappell, S.A., Wiegmann, D.A., 2001. Applying reason: the human factors analysis and classification system (HFACS). *Hum. Factors Aero Saf.* 1 (1), 59–86.
- Sterman, John D., 2000. *Business Dynamics: Systems Thinking and Modeling for a Complex World*, International student edition. McGraw-Hill, Irwin.
- Stewart, M.P., Martin, S.T., 2021. Unmanned aerial vehicles: fundamentals, components, mechanics, and regulations. In: Barrera, Nicholas (Ed.), *Unmanned Aerial Vehicles*. Nova Science Publishers.
- Stolzer, Alan J., Halford, Carl D., Goglia, John Joseph, 2011. *Implementing Safety Management Systems in Aviation*. Ashgate.
- Svedung, I., Rasmussen, J., 2002. Graphic representation of accidentscenarios: mapping system structure and the causation of accidents. *Saf. Sci.* 40 (5), 397–417. [https://doi.org/10.1016/S0925-7535\(00\)00036-9](https://doi.org/10.1016/S0925-7535(00)00036-9).
- Thompson, William T., Tvaryanas, Anthony P., Constable, Stefan H., 2005. *U.S. Military Unmanned Aerial Vehicle Mishaps: Assessment Of the Role of Human Factors Analysis and Classification System (HFACS)* (No. HSW-PE-BR-TR-2005-0001). Performance Enhancement Research Division, United States Air Force.
- Transport Safety Investigation Act 2003, 2024. Compilation No. 9, act No. 18, 2003 C2024C00120 (C10), 18, 2003. <https://www.legislation.gov.au/C2004A01102/la/test/text>.
- Tvaryanas, Anthony P., Thompson, William T., 2008. Recurrent error pathways in HFACS data: analysis of 95 mishaps with remotely piloted aircraft. *Aviat Space Environ. Med.* 79 (5), 525–532. <https://doi.org/10.3357/ASEM.2002.2008>.
- Tvaryanas, Anthony P., Thompson, William T., Constable, Stefan H., 2006. Human factors in remotely piloted aircraft operations: HFACS analysis of 221 mishaps over 10 years. *Aviat Space Environ. Med.* 77 (7), 724–732.
- Underwood, Peter, Waterson, Patrick, 2014. Systems thinking, the Swiss Cheese Model and accident analysis: a comparative systemic analysis of the Grayrigg train derailment using the ATSB, AcciMap and STAMP models. *Accid. Anal. Prev.* 68, 75–94. <https://doi.org/10.1016/j.aap.2013.07.027>.
- Vicente, Kim J., Christoffersen, Klaus, 2006. The Walkerton *E. coli* outbreak: a test of Rasmussen's framework for risk management in a dynamic society. *Theor. Issues Ergon. Sci.* 7 (2), 93–112. <https://doi.org/10.1080/14639220500078153>.
- Walden, David D., Roedler, Garry J., Forsberg, Kevin, Hamelin, R. Douglas, Shortell, Thomas M. (Eds.), 2015. *INCOSE Systems Engineering Handbook: A Guide for System Life Cycle Processes and Activities*, fourth ed. Wiley.
- Wasserman, S., Faust, K., 1994. *Social Network Analysis: Methods and Applications*. Cambridge University Press.
- Waterson, P., Jenkins, D.P., Salmon, P.M., Underwood, P., 2017. 'Remixing Rasmussen'. The evolution of Accimaps within systemic accident analysis. *Applied Ergonomics* 59 (PtB), 483–503. <https://doi.org/10.1016/j.apergo.2016.09.004>.
- Wiedemann, Mirjam, Liang, Man, Keremane, Ganesh, Quigley, Kate, 2024. Advanced Air Mobility: a comparative review of policies from around the world—lessons for Australia. *Transp. Res. Interdiscip. Perspect.* 24, 100988. <https://doi.org/10.1016/j.trip.2023.100988>.
- Wild, Graham, Gavin, Kellie, Murray, John, Silva, Jose, Baxter, Glenn, 2017. A post-accident analysis of civil remotely-piloted aircraft system accidents and incidents. *J. Aero. Technol. Manag.* 9 (2), 157–168. <https://doi.org/10.5028/jatm.v9i2.701>.
- Wild, Graham, Murray, John, Baxter, Glenn, 2016. Exploring civil drone accidents and incidents to help prevent potential air disasters. *Aerospace* 3 (3), 22. <https://doi.org/10.3390/aerospace3030022>.
- Zhang, Yingyu, Jing, Linlin, Sun, Chang, 2018. Systems-based analysis of China-tianjin port fire and explosion: a comparison of HFACS, AcciMap, and STAMP. *J. Fail. Anal. Prev.* 18 (6), 1386–1400. <https://doi.org/10.1007/s11668-018-0534-1>.